

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination 2024-25 (NEP)

DEPARTMENT OF GEOLOGY**Paper Code: ESMJ21 / ESMN21 — Mineral and Earth Material**

Time: 2:30 Hours

Maximum Marks: 70

Ref Code: CECNF/N/2024-25/26

*Instruction: Write your Roll No. at the top immediately on the receipt of this question paper.**Note: Answer two questions each from Section-A and Section-B. Section-C is compulsory. Figures in the right-hand margin indicate marks.***SECTION – A****Question 1**

[14 Marks]

Provide a detailed explanation of the optical properties of anisotropic minerals observed under both Plane-Polarized Light (PPL) and Cross-Polarized Light (XPL), accompanied by appropriate diagrams.

Question 2

[14 Marks]

What are the diagnostic optical properties of plagioclase feldspar? Two orthoclase feldspar crystals (gemstone: 'Moonstone') were excavated from two pegmatite veins at different locations in an Indian craton. Do they possess the same interfacial angles? Justify your answer.

Question 3

[14 Marks]

"When an external force is applied to a mineral, the mineral responds." Describe this response in terms of any two physical properties of a mineral.

Question 4

[14 Marks]

Augite and Hornblende both possess prismatic crystal faces. Do they belong to the same crystallographic system? Differentiate between these two minerals based on their diagnostic physical and optical properties.

SECTION – B**Question 5**

[14 Marks]

Explain / Answer in brief with suitable examples ($4 \times 3.5 = 14$ Marks):

- Why 5-fold, 7-fold, and 8-fold rotation axes are not possible in crystals?
- Explain the function of a reflecting goniometer and its uses.
- How would you differentiate between a closed form and an open form?
- Deduce the Miller indices from the following Weiss parameters:

(e) 1a, 2b, 3c

(f) 2a, 4b, ∞ c

Question 6

[14 Marks]

Answer the following with suitable diagrams and examples where necessary ($4 \times 3.5 = 14$ Marks):

- (a) Differentiate between dispersion and diffusion of light.
- (b) What is the function of the aperture diaphragm in a polarizing microscope?
- (c) Why is Canada balsam commonly used as an adhesive in the preparation of thin sections in petrography?
- (d) How is a Nicol prism constructed and what is its function in optical mineralogy?

Question 7

[14 Marks]

Answer the following with suitable examples where necessary ($4 \times 3.5 = 14$ Marks):

- (a) What is the basis for the classification of silicate minerals? Name the different classes of silicates.
- (b) Why do diamond and graphite have the same chemical composition but different specific gravity?
- (c) Why is the chemistry-based classification considered the most preferred classification scheme of minerals?
- (d) Discuss the igneous and sedimentary processes of mineral formation.

SECTION – C (Compulsory)**Question 8**

[14 Marks]

Define / Answer the following in one word or sentence ($14 \times 1 = 14$ Marks):

- (a) Minerals which are always black in crossed polars are called _____.
- (b) Which mineral shows polysynthetic twinning under the microscope?
- (c) What is the role of the auxiliary condensing lens?
- (d) How to differentiate between biotite and muscovite under the microscope?
- (e) If you are given a task to distinguish a mineral and an amorphous substance, how will it be done?
- (f) Name a non-ferromagnesian phyllosilicate mineral.
- (g) What are the composition and hardness of a streak plate?
- (h) What is the typical crystal form of garnet?
- (i) If a mineral can scratch fluorite but not quartz, its hardness on the Mohs scale is most likely _____.

- (j) _____ is an imaginary three-dimensional framework representing positions of motifs in a crystal.
- (k) A point where three or more crystal faces meet is called a _____.
- (l) Define pleochroic halo.
- (m) Define birefringence in minerals.
- (n) Give an example of an oxide mineral belonging to the spinel group.